

CSE401: Numerical Analysis, Simulation and Modeling

Root Finding: Bisection, False-Position & Newton–Raphson Method

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A situationship with $f(x) = 0$

- Find an x such that $f(x) = 0$.
- Not all equations have tidy closed-form root.
- Transcendental equations ($e^x = 3x$, $\sin x = x/2$, ...) decline to cooperate at all.
- So we approximate — iteratively — until the error is small enough.

The running example

All three methods will be tested on the **same unfortunate rubber ball**.

The problem

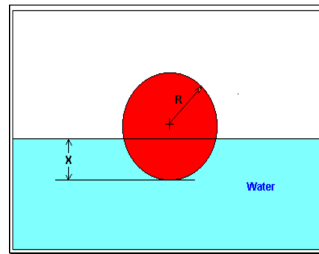
You drop a **rubber ball** into a tank of water, where it bobs contentedly at the surface, declining to sink. The ball has specific gravity 0.6 and radius 5.5 cm. Find the depth x to which it sits submerged.

The depth x (in metres) satisfies

$$f(x) = x^3 - 0.165x^2 + 3.993 \times 10^{-4} = 0.$$

x is bounded between the top and the bottom of the ball:

$$0 \leq x \leq 2R = 2(0.055) = 0.11.$$



Bisection Method

The big idea

- Bisection is like binary search.
- If a **continuous** function is **negative somewhere** and **positive somewhere else**, it had to cross zero in between.

The theorem it rests on

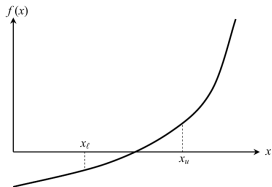
Theorem

If $f(x)$ is a real, continuous function and

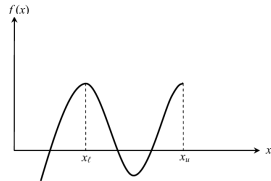
$$f(x_\ell) f(x_u) < 0,$$

then $f(x) = 0$ has **at least one** root between x_ℓ and x_u .

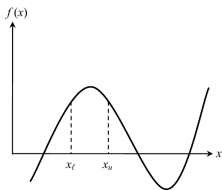
- If $f(x_\ell)f(x_u) > 0$, there **may or may not** be a root in between.
- If $f(x_\ell)f(x_u) < 0$, there could even be **more than one**.
- The theorem promises “at least one,” never “exactly one.”



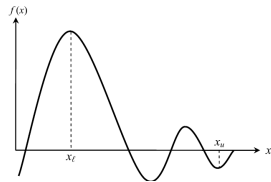
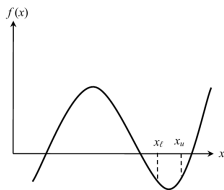
sign change, one root



No sign change, multiple roots



No sign change, no root



Sign change with multiple roots

The sign-change test is sufficient, not exhaustive. It tells you a root *exists*; but stays quiet about how many.

Halving the interval

- Take the midpoint $x_m = \frac{x_\ell + x_u}{2}$. This splits the bracket into $[x_\ell, x_m]$ and $[x_m, x_u]$.
- Which half holds the root? Check the sign of $f(x_\ell) f(x_m)$:
 - if $f(x_\ell)f(x_m) < 0$, the root is in the **left** half $[x_\ell, x_m]$;
 - otherwise it is in the **right** half $[x_m, x_u]$.
- Keep that half, repeat. We discard half the universe at every step.

Algorithm

1. Choose x_ℓ, x_u with $f(x_\ell)f(x_u) < 0$ (i.e. a sign change).
2. Estimate the root: $x_m = \frac{x_\ell + x_u}{2}$.
3. Decide the new bracket:
 - if $f(x_\ell)f(x_m) < 0$: root in $[x_\ell, x_m]$, so set $x_u = x_m$;
 - if $f(x_\ell)f(x_m) > 0$: root in $[x_m, x_u]$, so set $x_\ell = x_m$;
 - if $f(x_\ell)f(x_m) = 0$: x_m is the root — stop.
4. Compute new x_m and the **absolute relative approximate error** (look before you leap):

$$|\epsilon_a| = \left| \frac{x_m^{\text{new}} - x_m^{\text{old}}}{x_m^{\text{new}}} \right| \times 100\%.$$

5. If $|\epsilon_a| > \epsilon_s$ go back to step 3; else stop. (Also bail out if the iteration count gets out of hand.)

Two bookkeeping rules

Stopping error

$$|\epsilon_a| = \left| \frac{x_m^{\text{new}} - x_m^{\text{old}}}{x_m^{\text{new}}} \right| \times 100\% \text{ — compared each step against a tolerance } \epsilon_s.$$

Significant digits

At least m significant digits are correct once

$$|\epsilon_a| \leq 0.5 \times 10^{2-m}.$$

On the very first iteration there is no previous estimate, so ϵ_a is undefined

Let's start

Take $x_\ell = 0$ and $x_u = 0.11$ ($R = 0.055$). Check the sign change:

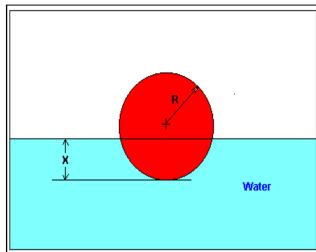
$$f(x_\ell) = f(0) = (0)^3 - 0.165(0)^2 + 3.993 \times 10^{-4} = +3.993 \times 10^{-4},$$

$$f(x_u) = f(0.11) = (0.11)^3 - 0.165(0.11)^2 + 3.993 \times 10^{-4} = -2.662 \times 10^{-4}.$$

Therefore

$$f(x_\ell) f(x_u) = (+3.993 \times 10^{-4})(-2.662 \times 10^{-4}) < 0.$$

A root is guaranteed somewhere in $[0, 0.11]$. Off we go.



Iteration 1

$$x_m = \frac{0 + 0.11}{2} = 0.055$$

$$f(0.055) = +6.655 \times 10^{-5}$$

$$f(x_\ell)f(x_m) = (+)(+) > 0$$

$\Rightarrow$ root in right half. New bracket: $[0.055, 0.11]$.

ϵ_a : not yet defined.



Iteration 2

$$x_m = \frac{0.055 + 0.11}{2} = 0.0825$$

$$f(0.0825) = -1.622 \times 10^{-4}$$

$$f(x_\ell)f(x_m) = (+)(-) < 0$$

$\Rightarrow$ root in left half. New bracket: $[0.055, 0.0825]$.

$$|\epsilon_a| = \left| \frac{0.0825 - 0.055}{0.0825} \right| \times 100 = 33.33\%$$



bracket $[0.055, 0.11]$

The bracket keeps shrinking. Slowly. Honestly.

Iteration 3

$$x_m = \frac{0.055 + 0.0825}{2} = 0.06875$$

$$f(0.06875) = -5.563 \times 10^{-5}$$

$$f(x_\ell)f(x_m) = (+)(-) < 0$$

$\Rightarrow$ root in left half. New bracket: $[0.055, 0.06875]$.

$$|\epsilon_a| = \left| \frac{0.06875 - 0.0825}{0.06875} \right| \times 100 = 20\%$$



bracket $[0.055, 0.0825]$

The bracket keeps shrinking. Slowly. Honestly.

Iteration 4

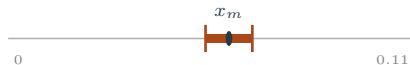
$$x_m = \frac{0.055 + 0.06875}{2} = 0.06188$$

$$f(0.06188) = +4.484 \times 10^{-6}$$

$$f(x_\ell)f(x_m) = (+)(+) > 0$$

$\Rightarrow$ root in right half. New bracket: $[0.06188, 0.06875]$.

$$|\epsilon_a| = \left| \frac{0.06188 - 0.06875}{0.06188} \right| \times 100 = 11.11\%$$



bracket $[0.055, 0.06875]$

The bracket keeps shrinking. Slowly. Honestly.

Iteration 5

$$x_m = \frac{0.06188 + 0.06875}{2} = 0.06531$$

$$f(0.06531) = -2.593 \times 10^{-5}$$

$$f(x_\ell)f(x_m) = (+)(-) < 0$$

$\Rightarrow$ root in left half. New bracket: $[0.06188, 0.06531]$.

$$|\epsilon_a| = \left| \frac{0.06531 - 0.06188}{0.06531} \right| \times 100 = 5.263\%$$



bracket $[0.06188, 0.06875]$

The bracket keeps shrinking. Slowly. Honestly.

Iter	x_ℓ	x_u	x_m	$f(x_m)$	$\text{sgn}[f_\ell, f_m, f_u]$	$ \epsilon_a \%$
1	0.00000	0.11	0.055	6.655×10^{-5}	+ + -	—
2	0.055	0.11	0.0825	-1.622×10^{-4}	+ - -	33.33
3	0.055	0.0825	0.06875	-5.563×10^{-5}	+ - -	20.00
4	0.055	0.06875	0.06188	4.484×10^{-6}	+ + -	11.11
5	0.06188	0.06875	0.06531	-2.593×10^{-5}	+ - -	5.263
6	0.06188	0.06531	0.06359	-1.0804×10^{-5}	+ - -	2.702
7	0.06188	0.06359	0.06273	-3.176×10^{-6}	+ - -	1.370
8	0.06188	0.06273	0.06230	6.497×10^{-7}	+ + -	0.6897
9	0.06230	0.06273	0.06252	-1.265×10^{-6}	+ - -	0.3436
10	0.06230	0.06252	0.06241	-3.0768×10^{-7}	+ - -	0.1721

How many digits can we trust?

After the 10th iteration, $|\epsilon_a| = 0.1721\%$. Find the largest m with $|\epsilon_a| \leq 0.5 \times 10^{2-m}$:

$$0.1721 \leq 0.5 \times 10^{2-m}$$

$$0.3442 \leq 10^{2-m}$$

$$\log(0.3442) \leq 2 - m$$

$$m \leq 2 - \log(0.3442) = 2.463 \Rightarrow m = 2.$$

So the estimate $x_m = 0.06241$ has **at least 2 correct significant digits**. The ball sits about 6.2 cm deep, for the curious.

Advantages

- **Always convergent.** The bisection method is always convergent. Since the method brackets the root, the method is guaranteed to converge.
- **Predictable error.** As iterations are conducted, the interval gets halved. So one can guarantee the error in the solution of the equation. So you can compute, in advance, how many iterations you will need.

Drawbacks

- **Slow convergence.** Simply based on halving the interval.
- **A good guess is wasted.** Even if one endpoint sits right on top of the root, bisection ignores that and marches to the midpoint anyway. Only sign changes, no proximity.

Drawbacks

Roots that merely touch the axis.

For $f(x) = x^2 = 0$ the curve touches the axis at $x = 0$ without changing sign.

Unable to find the lower guess x_ℓ and upper guess x_u such that $f(x_\ell)f(x_u) < 0$.

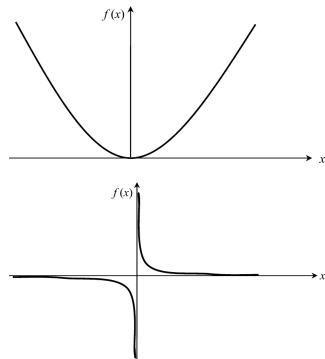
Singularities that flip sign.

For $f(x)$ which reverses sign at singularity, bisection may converge at singularity.

Ex. $f(x) = \frac{1}{x}$ the guesses $x_\ell = -2$, $x_u = 3$ satisfy $f(x_\ell)f(x_u) < 0$

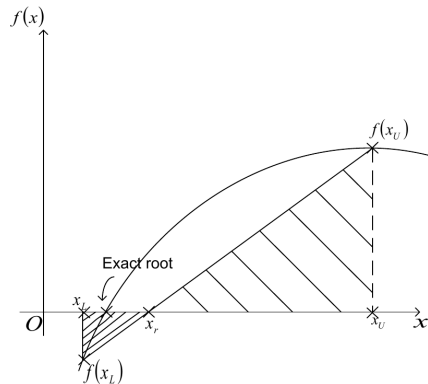
But *no* root. The function is discontinuous. Converges at the singularity.

A singularity in a function is defined as a point where the function becomes infinite.



False-Position Method

- Bisection blindly takes the **midpoint**, ignoring how close each endpoint's function value is to zero.
- But if $f(x_\ell)$ is much nearer zero than $f(x_u)$, the root is surely nearer x_ℓ than the midpoint.
- **False-position** draws a straight **secant** from $(x_\ell, f(x_\ell))$ to $(x_u, f(x_u))$ and takes where it crosses the axis as the new estimate.



Derivation of the formula

From the two similar triangles in the figure,

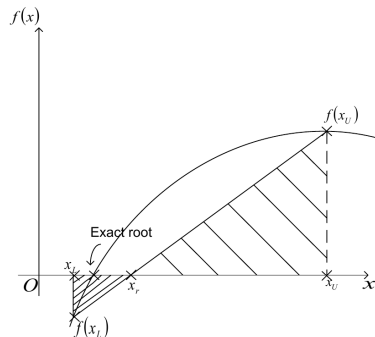
$$\frac{0 - f(x_L)}{x_r - x_L} = \frac{0 - f(x_U)}{x_r - x_U}.$$

Cross-multiplying and tidying up:

$$x_U f(x_L) - x_L f(x_U) = x_r \{f(x_L) - f(x_U)\}.$$

Solving for the new estimate x_r :

$$x_r = \frac{x_U f(x_L) - x_L f(x_U)}{f(x_L) - f(x_U)}$$



Three faces of the same formula

The estimate can be rewritten in two “increment” forms:

$$x_r = x_U - \frac{f(x_U)(x_U - x_L)}{f(x_U) - f(x_L)} \quad \text{or} \quad x_r = x_L - \frac{f(x_L)(x_L - x_U)}{f(x_L) - f(x_U)}.$$

Algorithm

1. Choose x_L, x_U with $f(x_L)f(x_U) < 0$.
2. Estimate the root: $x_r = x_U - \frac{f(x_U)(x_U - x_L)}{f(x_U) - f(x_L)}$ or $x_r = x_L - \frac{f(x_L)(x_L - x_U)}{f(x_L) - f(x_U)}$.
3. Decide the new bracket:
 - if $f(x_L)f(x_r) < 0$: root in $[x_L, x_r]$, so set $x_U = x_r$;
 - if $f(x_L)f(x_r) > 0$: root in $[x_r, x_U]$, so set $x_L = x_r$;
 - if $f(x_L)f(x_r) = 0$: x_r is the root — stop.
4. Compute x_r and calculate the **absolute relative approximate error**:
$$|\epsilon_a| = \left| \frac{x_r^{\text{new}} - x_r^{\text{old}}}{x_r^{\text{new}}} \right| \times 100\%.$$
5. If $|\epsilon_a| > \epsilon_s$ go back to step 3; else stop. (Also bail out if the iteration count gets out of hand.)

The only difference from bisection is the formula in steps 2 and 4. The rest is the same.

Same ball, same setup

Take $x_L = 0$, $x_U = 0.11$. The function values are exactly as before:

$$f(x_L) = +3.993 \times 10^{-4}, \quad f(x_U) = -2.662 \times 10^{-4},$$

so $f(x_L)f(x_U) < 0$ and a root lives in $[0, 0.11]$.

Iteration 1

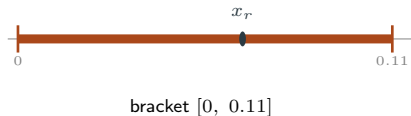
$$x_r = \frac{0.11(3.993 \times 10^{-4}) - 0}{3.993 \times 10^{-4} - (-2.662 \times 10^{-4})} = 0.0660$$

$$f(0.0660) = -3.1944 \times 10^{-5}$$

$$f(x_L)f(x_r) = (+)(-) < 0$$

$\Rightarrow$ new bracket $[0, 0.0660]$.

ϵ_a : not yet defined.



Iteration 2

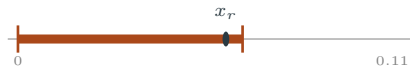
$$x_r = \frac{0.0660(3.993 \times 10^{-4}) - 0}{3.993 \times 10^{-4} - (-3.1944 \times 10^{-5})} = 0.0611$$

$$f(0.0611) = +1.1320 \times 10^{-5}$$

$$f(x_L)f(x_r) = (+)(+) > 0$$

$\Rightarrow$ new bracket $[0.0611, 0.0660]$.

$$|\epsilon_a| = \left| \frac{0.0611 - 0.0660}{0.0611} \right| \times 100 \approx 8\%$$



bracket $[0, 0.0660]$

Iteration 3

$$x_r = \frac{0.0660(1.132 \times 10^{-5}) - 0.0611(-3.1944 \times 10^{-5})}{1.132 \times 10^{-5} - (-3.1944 \times 10^{-5})} = 0.0624$$

$$f(0.0624) = -1.1313 \times 10^{-7}$$

$$f(x_L)f(x_r) = (+)(-) < 0$$

$\Rightarrow$ new bracket $[0.0611, 0.0624]$.

$$|\epsilon_a| = \left| \frac{0.0624 - 0.0611}{0.0624} \right| \times 100 \approx 2.05\%$$



bracket $[0.0611, 0.0660]$

Watch $f(x_r)$: $10^{-5} \rightarrow 10^{-7} \rightarrow 10^{-13}$.

Meanwhile x_L sits frozen at 0.0611.

Iteration 4

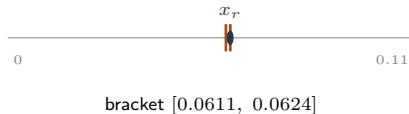
$$x_r = \frac{0.0624(1.132 \times 10^{-5}) - 0.0611(-1.1313 \times 10^{-7})}{1.132 \times 10^{-5} - (-1.1313 \times 10^{-7})} = 0.062378$$

$$f(0.062378) = -3.347 \times 10^{-10}$$

$$f(x_L)f(x_r) = (+)(-) < 0$$

$\Rightarrow$ new bracket $[0.0611, 0.06238]$.

$|\epsilon_a| \approx 0.020\%$ (x_L still stuck at 0.0611).



Watch $f(x_r)$: $10^{-5} \rightarrow 10^{-7} \rightarrow 10^{-13}$.

Meanwhile x_L sits frozen at 0.0611.

Iteration 5

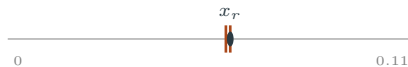
$$x_r = \frac{0.06238(1.132 \times 10^{-5}) - 0.0611(-3.347 \times 10^{-10})}{1.132 \times 10^{-5} - (-3.347 \times 10^{-10})} = 0.062378$$

$$f(0.062378) = -9.896 \times 10^{-13}$$

$$f(x_L)f(x_r) = (+)(-) < 0$$

$\Rightarrow$ new bracket $[0.0611, 0.06238]$.

$|\epsilon_a| \approx 1 \times 10^{-4}\%$ (x_r has stopped moving; f keeps falling).



bracket $[0.0611, 0.06238]$

Watch $f(x_r)$: $10^{-5} \rightarrow 10^{-7} \rightarrow 10^{-13}$.

Meanwhile x_L sits frozen at 0.0611.

Iter	x_L	x_U	x_r	$f(x_r)$	$\text{sgn}[f_L, f_r, f_U]$	$ \epsilon_a \%$
1	0.0000	0.1100	0.0660	-3.1944×10^{-5}	+ - -	—
2	0.0000	0.0660	0.0611	$+1.1320 \times 10^{-5}$	+ + -	8.00
3	0.0611	0.0660	0.0624	-1.1313×10^{-7}	+ - -	2.05
4	0.0611	0.0624	0.06238	-3.347×10^{-10}	+ - -	0.020
5	0.0611	0.06238	0.06238	-9.896×10^{-13}	+ - -	0.0001

The sign column lists $\text{sgn } f$ at x_L, x_r, x_U : the root sits where the sign flips.

By iteration 5, $|\epsilon_a| \approx 10^{-4}\%$ (≈ 5 correct digits) and $f(x_r) \sim 10^{-13}$; bisection at the same count was still near 10^{-5} .

Yet another example

Find the root of $f(x) = (x - 4)^2(x + 2) = 0$ with $x_L = -2.5$, $x_U = -1.0$, tolerance $\epsilon_s = 0.1\%$.

Iter	x_L	x_U	$f(x_L)$	$f(x_U)$	x_r	$ \epsilon_a \%$
1	-2.5	-1.000	-21.13	25.000	-1.813	—
2	-2.5	-1.813	-21.13	6.319	-1.971	8.024
3	-2.5	-1.971	-21.13	1.028	-1.996	1.229
4	-2.5	-1.996	-21.13	0.1542	-1.999	0.1828
5	-2.5	-1.999	-21.13	0.02286	-2.000	0.02706

Five iterations reach the root $x = -2$. Significant digits: $0.02706 \leq 0.5 \times 10^{2-m} \Rightarrow m \leq 3.27$, so **at least 3 correct digits**.

Can False-Position Be Slower Than Bisection?

Test function: $f(x) = \ln x$, exact root $x^* = 1$

Setup

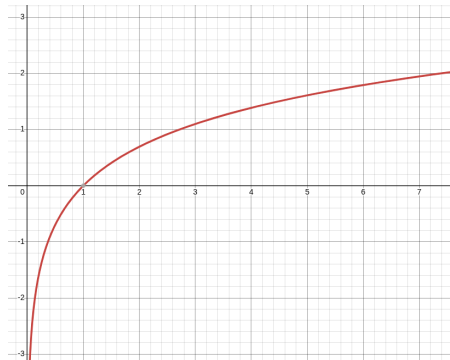
Bracket : $x_\ell = 10^{-4}$, $x_u = 10^4$

Stopping criterion : $|\epsilon_a| < 0.0001\%$

Compilation : g++ -O9

Results (averaged over 10 runs)

	Bisection	False-Position
Iterations	34	126
Wall time (ns)	5 418	14 753
Bisection was 2.7 × faster.		



Newton–Raphson Method

Open vs. bracketing methods

- Bisection and false-position need **two** guesses and a sign change. They always converge — but slowly.
- Newton–Raphson is an **open method**: it needs only **one** initial guess, no bracket.
- Convergence is **not guaranteed**. But when it does converge, it does **much faster** than any bracketing method.

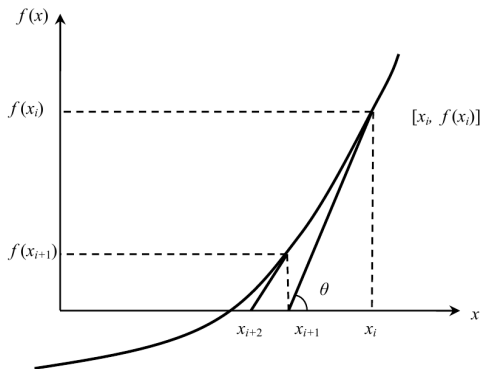
Derivation by tangent line

- Stand at x_i . Draw the **tangent** to the curve at $(x_i, f(x_i))$.
- Where that tangent crosses the x -axis is your next estimate x_{i+1} .
- The slope is the derivative:

$$f'(x_i) = \tan \theta = \frac{f(x_i) - 0}{x_i - x_{i+1}}.$$

- Solve for x_{i+1} :

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$



The entire method fits on one line.

Algorithm

1. Evaluate $f'(x)$.
2. From a guess x_i , compute $x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$.
3. Find the error $|\epsilon_a| = \left| \frac{x_{i+1} - x_i}{x_{i+1}} \right| \times 100\%$.
4. If $|\epsilon_a| > \epsilon_s$ go back to step 2; else stop (and watch the iteration cap).

Setting up the floating ball

$$f(x) = x^3 - 0.165x^2 + 3.993 \times 10^{-4}, \quad f'(x) = 3x^2 - 0.33x.$$

Take the initial guess $x_0 = 0.05$ m — a sensible spot between 0 and the diameter 0.11. **Why not $x_0 = 0$ or $x_0 = 0.11$?** Because

$$f'(x) = 3x(x - 0.11) \quad \Rightarrow \quad f'(0) = 0, \quad f'(0.11) = 0.$$

Both extreme guesses give a **horizontal tangent**; division by zero, and a flat line which never meets the axis.

Iteration 1 ($x_0 = 0.05$)

$$x_1 = 0.05 - \frac{(0.05)^3 - 0.165(0.05)^2 + 3.993 \times 10^{-4}}{3(0.05)^2 - 0.33(0.05)}$$

$$x_0 = 0.05$$

$$= 0.05 - \frac{1.118 \times 10^{-4}}{-9 \times 10^{-3}} = 0.05 - (-0.01242)$$

$$x_1 = 0.06242$$

$$= 0.06242$$

$$x_2 = 0.06238$$

$$|\epsilon_a| = \left| \frac{0.06242 - 0.05}{0.06242} \right| \times 100 = 19.90\% \quad (0 \text{ digits})$$

$$x_3 = 0.06238$$

Iteration 2 ($x_1 = 0.06242$)

$$x_2 = 0.06242 - \frac{3.97781 \times 10^{-7}}{8.90973 \times 10^{-3}}$$

$$= 0.06242 - 4.4646 \times 10^{-5} = 0.06238$$

$$|\epsilon_a| = \left| \frac{0.06238 - 0.06242}{0.06238} \right| \times 100 = 0.0716\% \quad (2 \text{ digits})$$

$$x_0 = 0.05$$



$$x_1 = 0.06242$$



$$x_2 = 0.06238$$

$$x_3 = 0.06238$$

Iteration 3 ($x_2 = 0.06238$)

$$x_3 = 0.06238 - \frac{4.44 \times 10^{-11}}{8.91171 \times 10^{-3}}$$

$$= 0.06238 - 4.9822 \times 10^{-9} = 0.06238$$

$|\epsilon_a| = 0\%$ (4 digits — only 4 were carried)

$$\begin{array}{c} x_0 = 0.05 \\ \downarrow \\ x_1 = 0.06242 \\ \downarrow \\ x_2 = 0.06238 \\ \downarrow \\ x_3 = 0.06238 \end{array}$$

Three iterations: error $19.9\% \rightarrow 0$. The digits roughly **doubled** each step here.

Drawbacks — divergence at inflection points

For $f(x) = (x - 1)^3 + 0.512 = 0$, Newton–Raphson reduces to

$$x_{i+1} = x_i - \frac{(x_i - 1)^3 + 0.512}{3(x_i - 1)^2}.$$

What is an inflection point?

- An **inflection point** is where the concavity flips: up-to-down or down-to-up.
- For $f(x) = (x - 1)^3$, concavity changes at $x = 1$, so $(1, 0)$ is an inflection point.
- $f''(x) = 0$ at inflection points, but it does not guarantee inflection.
- For $f(x) = x^4 - 16$, $f''(0) = 0$, but concavity does *not* flip, so $(0, -16)$ is **not** an inflection point.

Inflection Point

If $f'(c)$ exists and $f''(c)$ changes sign at $x = c$, then the point $(c, f(c))$ is an inflection point of the graph of f .

Drawbacks — divergence at inflection points

For $f(x) = (x - 1)^3 + 0.512 = 0$, Newton–Raphson reduces to

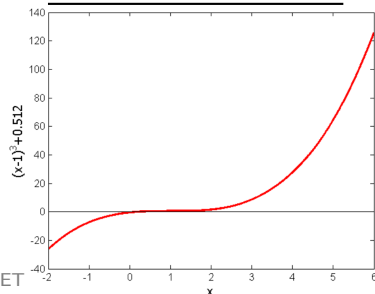
$$x_{i+1} = x_i - \frac{(x_i - 1)^3 + 0.512}{3(x_i - 1)^2}.$$

Start at $x_0 = 5.0$. The estimate 0.92589 at iteration 5 lands near the **inflection point** $x = 1$, where $f' = 0$.

It panics, sprints to -30.119 , reconsiders its life choices, and crawls back. After 12 more iterations it reaches $x = 0.2$.

A bracketing method would have boxed this in immediately.

i	x_i	i	x_i
0	5.0000	10	-5.1498
1	3.6560	11	-3.1044
2	2.7465	12	-1.7464
3	2.1084	13	-0.85356
4	1.6000	14	-0.28538
5	0.92589	15	0.039784
6	-30.119	16	0.17475
7	-19.746	17	0.19924
8	-12.831	18	0.2
9	-8.2217		



Drawbacks — division by zero (or nearly)

For $f(x) = x^3 - 0.03x^2 + 2.4 \times 10^{-6} = 0$,

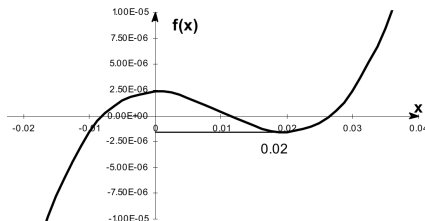
$$x_{i+1} = x_i - \frac{x_i^3 - 0.03x_i^2 + 2.4 \times 10^{-6}}{3x_i^2 - 0.06x_i}.$$

The denominator $3x_i^2 - 0.06x_i$ vanishes at $x = 0$ and $x = 0.02$.

Start near 0.02, say $x_0 = 0.01999$: you dodge an exact zero, but the denominator is microscopic.

The method launches off to nowhere; after 9 iterations it has *not* converged.

i	x_i	$f(x_i)$	$ \epsilon_a \%$
0	0.019990	-1.6×10^{-6}	—
1	-2.6480	18.778	100.75
2	-1.7620	-5.5638	50.282
3	-1.1714	-1.6485	50.422
4	-0.77765	-0.48842	50.632
5	-0.51518	-0.14470	50.946
6	-0.34025	-0.042862	51.413
7	-0.22369	-0.012692	52.107
8	-0.14608	-0.0037553	53.127
9	-0.094490	-0.0011091	54.602

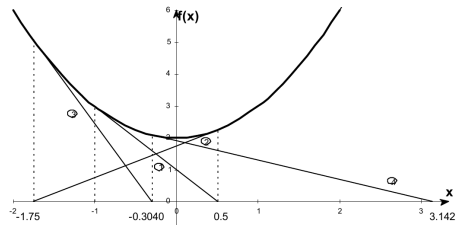


Drawbacks — oscillation near a max/min

For $f(x) = x^2 + 2 = 0$ — which has **no real roots** (there is your first problem) — the iterates bounce around the minimum forever.

They never settle, and eventually drift toward a near-zero slope and diverge.

i	x_i	$f(x_i)$	$ \epsilon_a \%$
0	-1.0000	3.00	—
1	0.5	2.25	300.00
2	-1.75	5.063	128.571
3	-0.30357	2.092	476.47
4	3.1423	11.874	109.66
5	1.2529	3.570	150.80
6	-0.17166	2.029	829.88
7	5.7395	34.942	102.99
8	2.6955	9.266	112.93
9	0.97678	2.954	175.96



Drawbacks — root jumping

For $f(x) = \sin x = 0$ there are infinitely many roots.

Start at $x_0 = 2.4\pi \approx 7.539822$, intending to land on $2\pi \approx 6.283$.

Instead it leaps across several roots and converges to $x = 0$.

You aim at one root $x = 2\pi$; it converges to a completely different one and offers no apology.

i	x_i	$f(x_i)$	$ \epsilon_a \%$
0	7.539822	0.951	—
1	4.462	-0.969	68.973
2	0.5499	0.5226	711.44
3	-0.06307	-0.06303	971.91
4	8.376×10^{-4}	8.375×10^{-4}	7.54×10^4
5	-1.95861×10^{-13}	-1.95861×10^{-13}	4.28×10^{10}

